G-8 Game code Platformer physics Tutorial

Save your work in scratch using scratch ID

ID: Linkscoders

password: Linkscoders123

Use outlook ID to send email to teacher

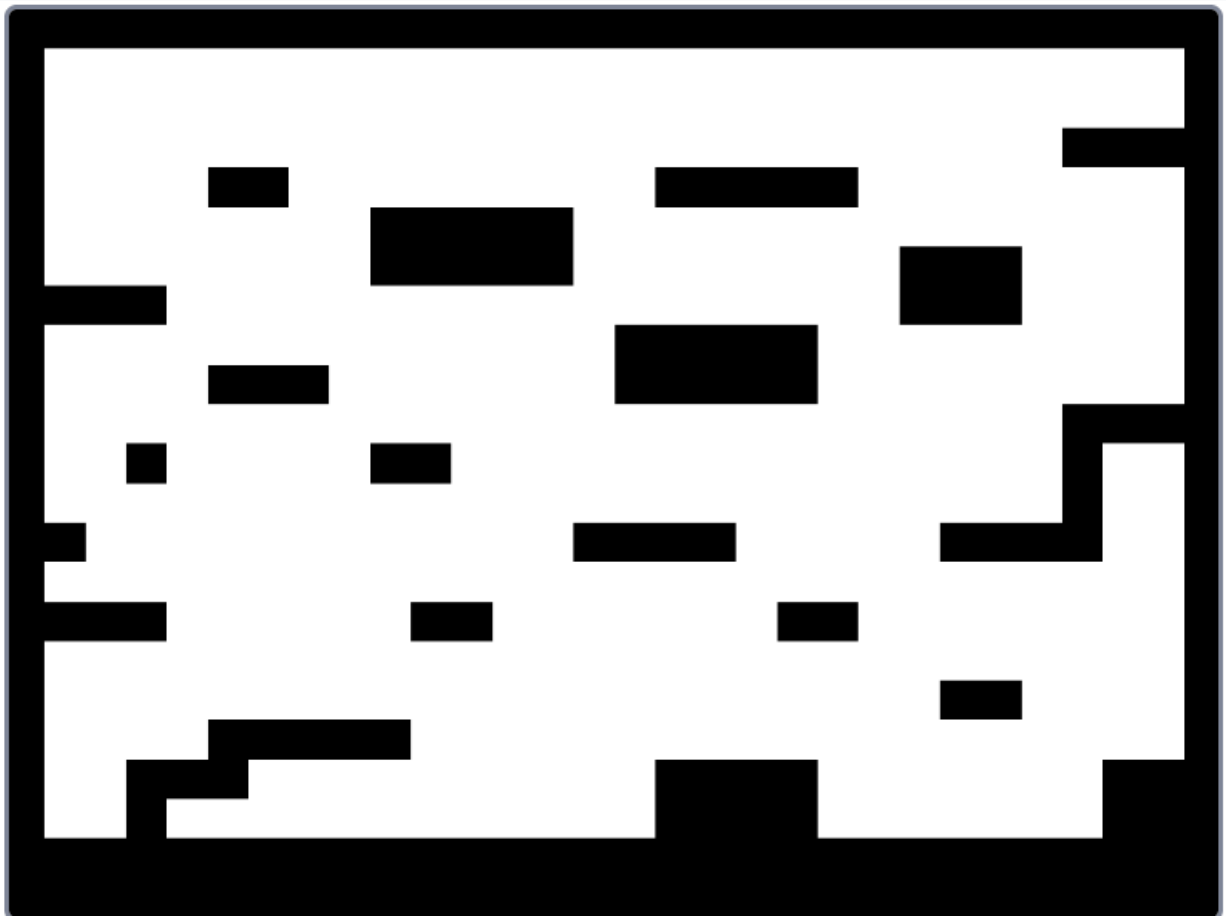
ID: Coding1@linksedu.com

Pass: Links2026

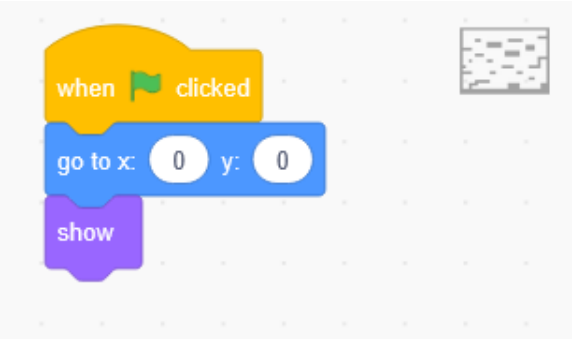
Email your sb3 file on **teachers Id**: Coding10@linksedu.com

Tutorial: https://www.youtube.com/watch?v=pPNrr6sAZaI

Background:

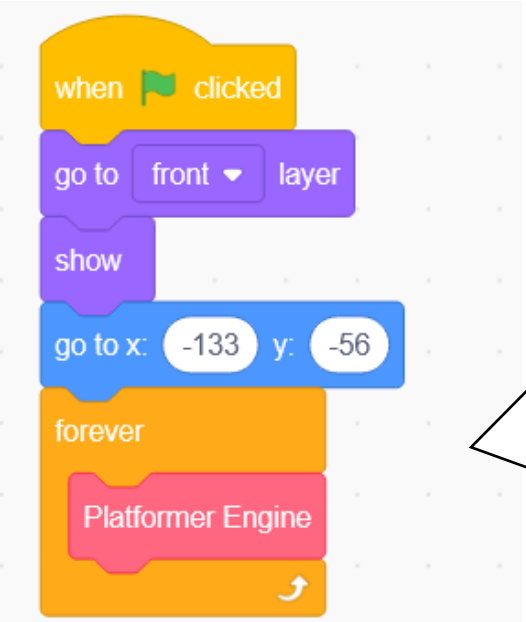


Code for background:



Code for the platformer:

Define position and introduce a customized block using my block option

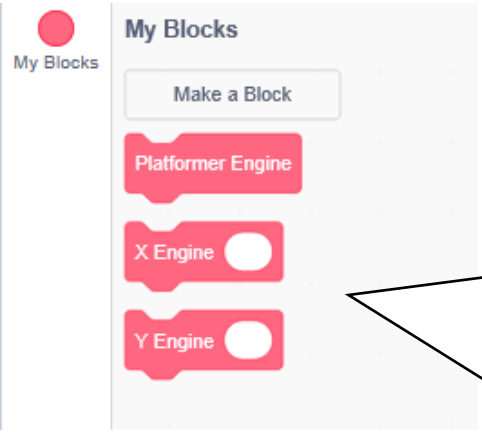


When Green Flag Clicked

- The sprite **moves to the starting position** (x: -133, y: -56).
- It **comes to the front layer** and becomes visible.
- Then it **runs the Platformer Engine forever**.

☞ This means the game keeps checking movement and physics all the time.

create the following blocks, tick the option given below

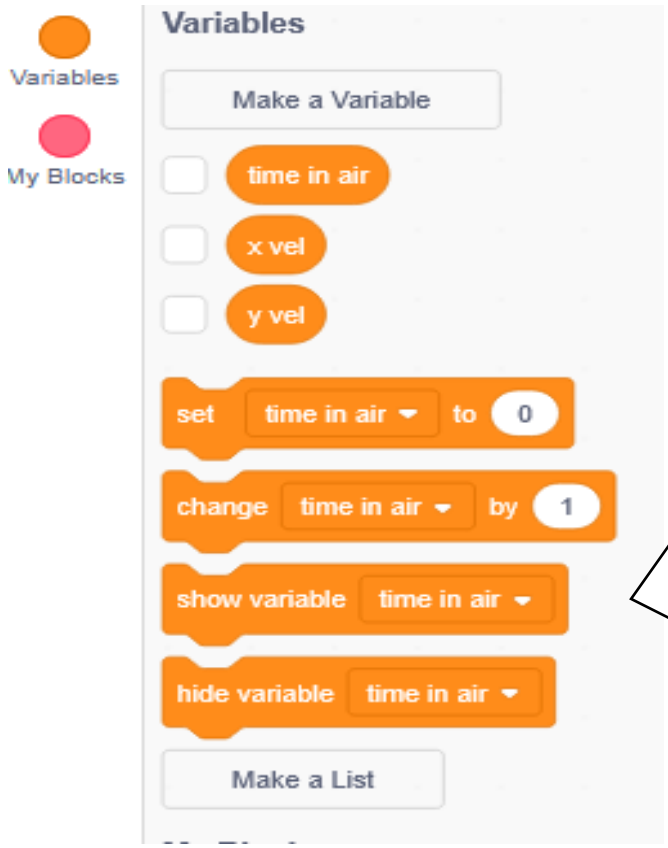


Main Parts of the Platformer Engine

Your engine is divided into **3 small systems**:

1. **X Engine (Horizontal Movement)**
2. **Y Engine (Vertical Movement / Gravity / Jumping)**
3. **Platformer Engine (Controls + Physics)**

Create following variables:

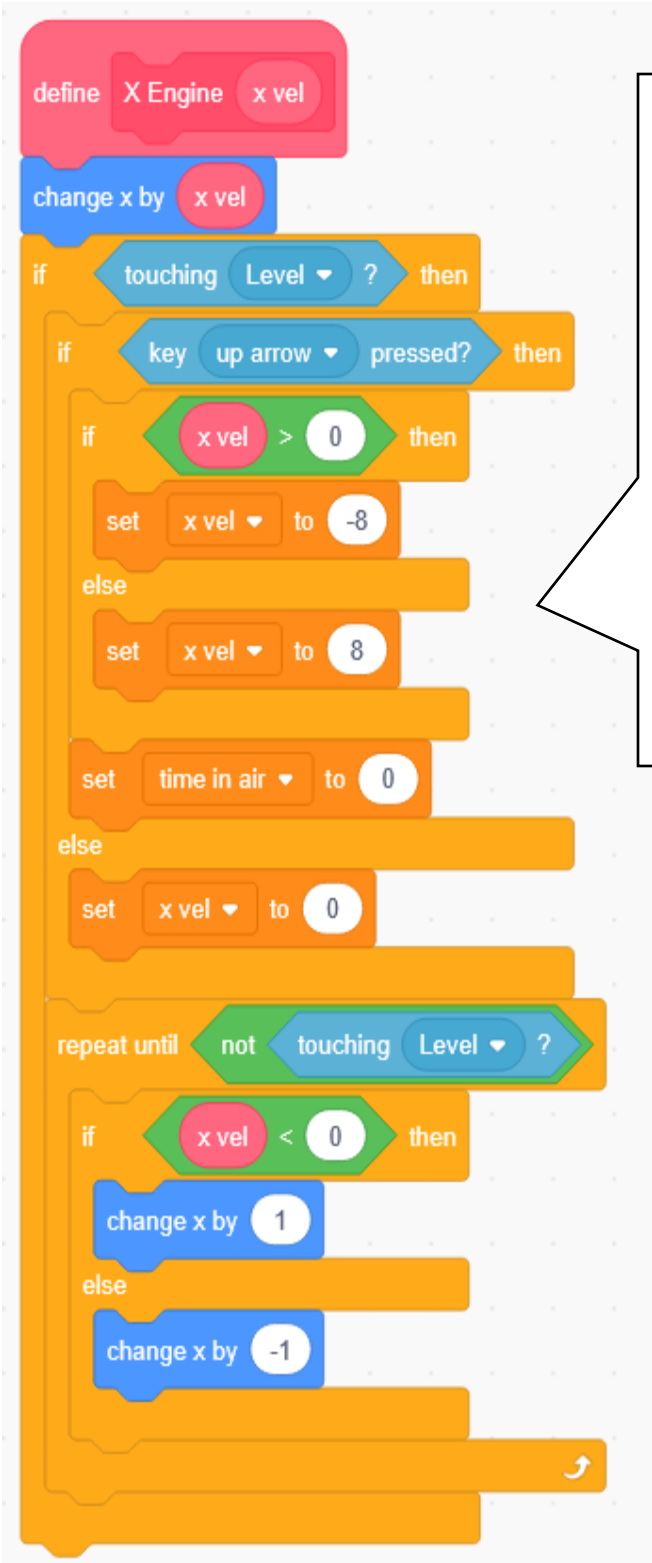


Variables Used

x vel
→ Player speed left/right.

y vel
→ Player speed up/down.

time in air
→ Counts how long the player is in the air (helps control jumping).



X Engine (Horizontal Collision)

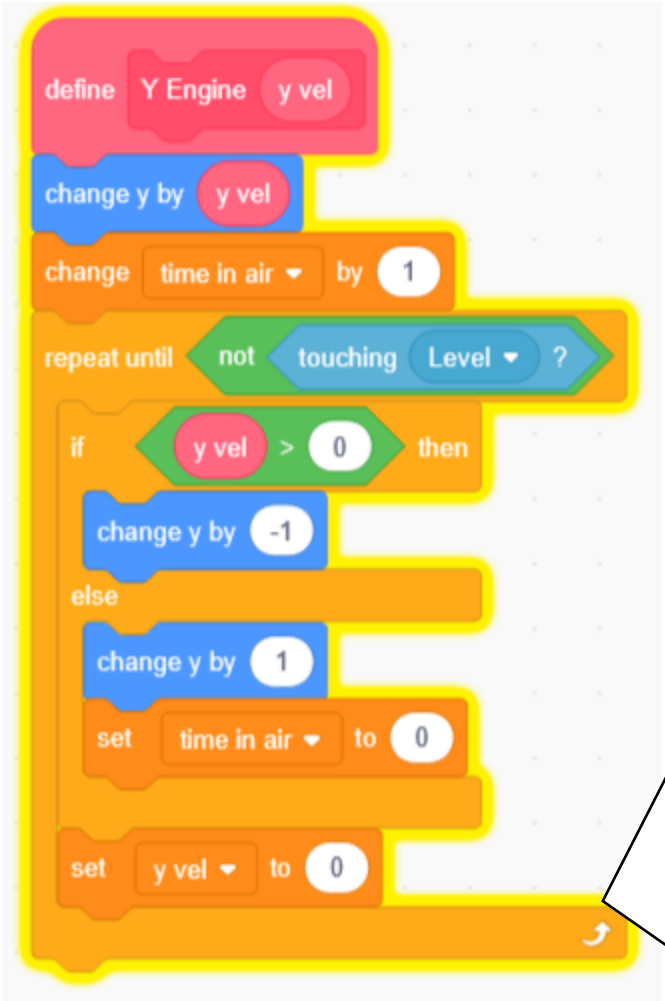
`change x by x vel`

→ Moves the player sideways.

If the sprite **touches the level (wall/platform)**:

- The code moves the sprite **back one step at a time**
- Until it is **no longer touching the wall**

→ This stops the player from going inside walls.



Y Engine (Vertical Movement)

```
change y by y vel
```

→ Moves the player up or down.

Gravity Timer

```
change time in air by 1
```

→ Counts how long the player is in the air.

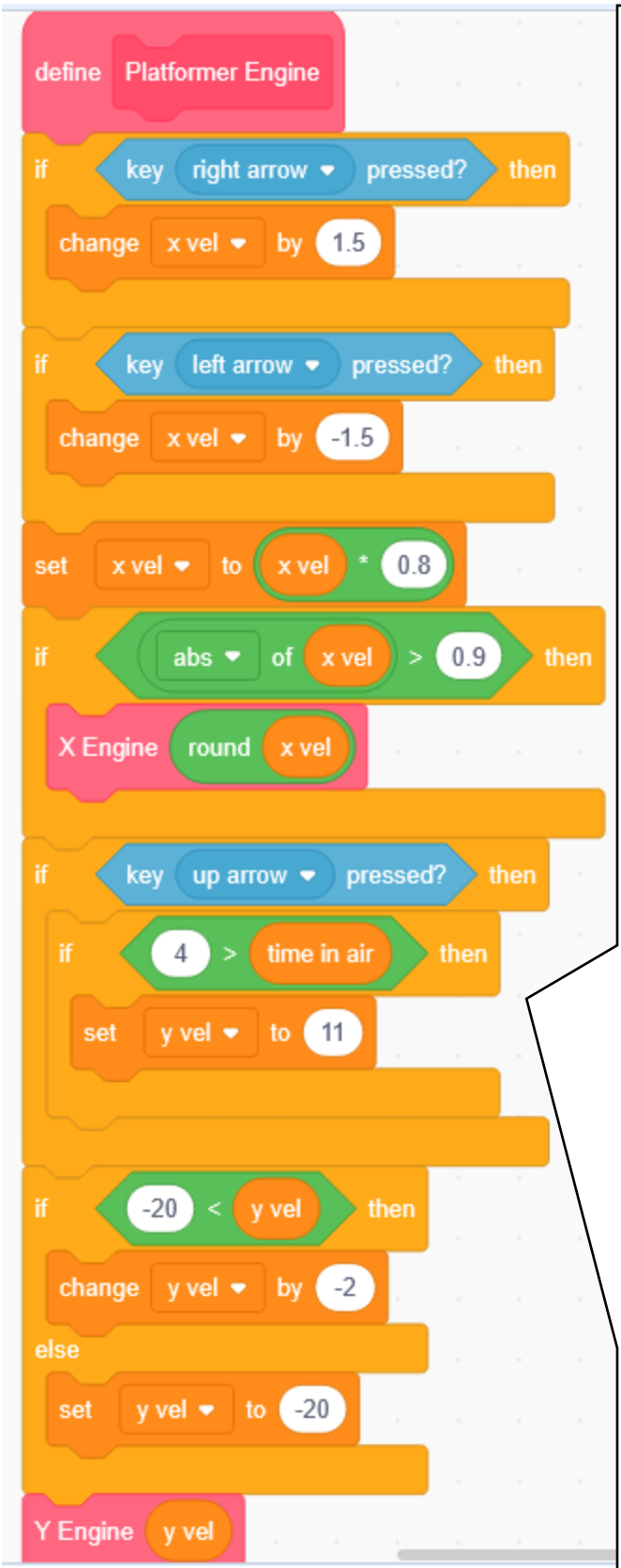
Ground Collision

If touching the ground:

- Move the sprite back until it is **not touching the ground**
- Set:

```
y vel = 0
time in air = 0
```

→ This makes the player **land on the platform correctly**.



Platformer Engine (Player Controls)

Move Right

- If **Right Arrow is pressed**
- Increase **x velocity (x vel)**

→ The character moves right.

Move Left

- If **Left Arrow is pressed**
- Decrease **x velocity**

→ The character moves left.

Friction

```
set x vel to x vel * 0.8
```

→ Slows the player down gradually when no key is pressed.

Jump

If:

- **Up Arrow pressed**
- Player is **on the ground (time in air < 4)**

Then:

- **y vel = 11**

→ The player jumps.

Gravity

```
change y vel by -2
```

→ Pulls the player downward.

Further detail of each step is shared below

This script **controls the player movement**, including:

- Moving left and right
- Jumping
- Gravity (falling down)

---

1 Move Right

Block:

```
if right arrow pressed
change x vel by 1.5
```

Meaning:

- When the **Right Arrow key** is pressed,
- The **horizontal speed (x vel)** increases.

✅ Result: The character **moves right**.

---

2 Move Left

Block:

```
if left arrow pressed
change x vel by -1.5
```

Meaning:

- When the **Left Arrow key** is pressed,
- The **horizontal speed decreases**.

✅ Result: The character **moves left**.

---

3 Friction (Slowing Down)

Block:

```
set x vel to x vel × 0.8
```

Meaning:

- The speed is multiplied by **0.8** every frame.

✅ Result:

- The character **slowly stops** when no key is pressed.
- This makes movement **smooth and realistic**.

---

4 Move the Player Horizontally

Block:

```
if abs of x vel > 0.9
X Engine (round x vel)
```

Meaning:

- If the horizontal speed is big enough,
- The **X Engine block moves the sprite**.

`abs` means **absolute value** (ignores negative sign).

✅ Result:

- The sprite moves left or right.

---

5 Jump

Block:

```
if up arrow pressed
if 4 > time in air
set y vel to 11
```

Meaning:

- If the **Up Arrow key** is pressed
- AND the player **just touched the ground**

Then:

- The vertical speed **y vel becomes 11**.

✅ Result: The character **jumps up**.

---

6 Gravity (Falling Down)

Block:

```
if -20 < y vel
change y vel by -2
```

Meaning:

- Gravity reduces the vertical speed.

Every frame:

- **y vel decreases by 2**

✅ Result:

- The character **falls down naturally**.

---

7 Falling Speed Limit

Block:

```
else
set y vel to -20
```

Meaning:

- The falling speed cannot go faster than **-20**.

✅ Result:

- Prevents the player from **falling too fast**.

---

8 Move Vertically

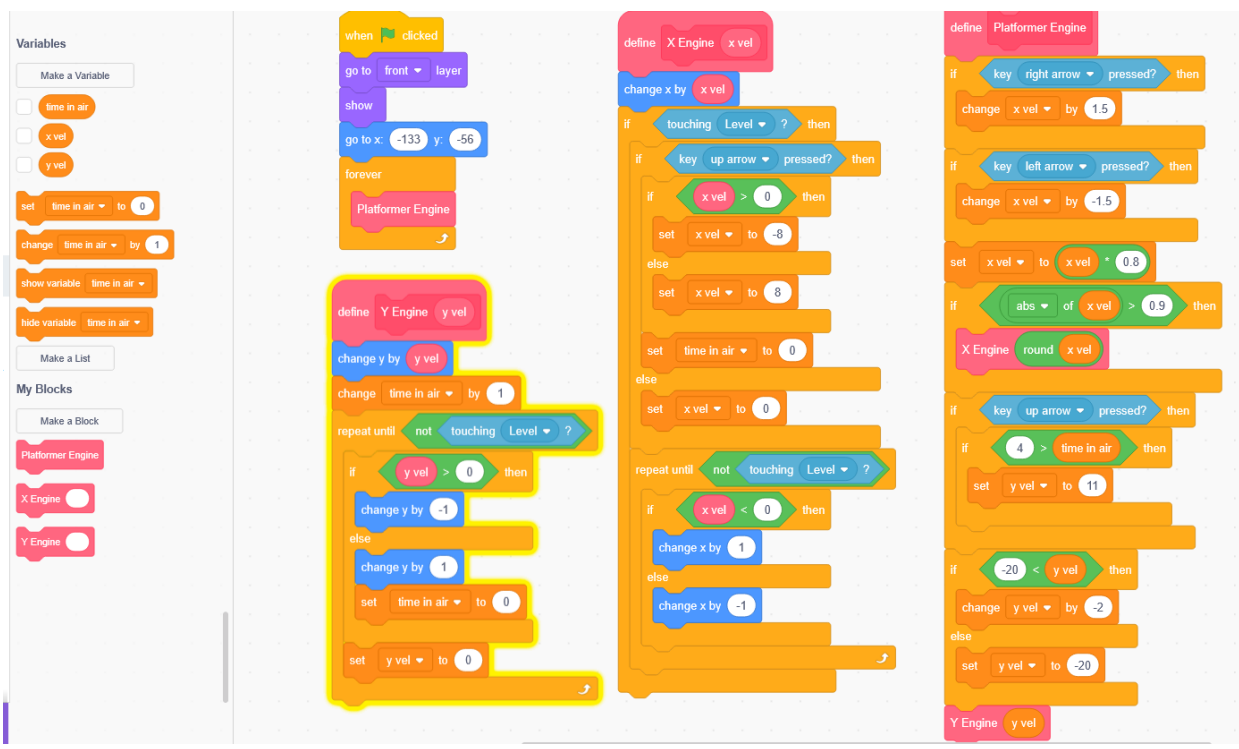
Block:

```
Y Engine (y vel)
```

Meaning:

- This block moves the player **up or down** using the vertical speed.

✅ Result:

- Jumping and falling works.

Final Look: